

TERRITORY GRAPE FARM, Australia

WMO: 943280

Lat: 22.4517S

Lon: 133.6378E

Elev: 567

StdP: 94.70

Time Zone: 9.50 (AUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	3.0	4.4	-13.6	1.2	21.0	-11.5	1.5	21.0	8.9	17.5	8.1	16.6	2.1	170	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	40.0	18.6	38.7	18.6	37.7	18.7	29.2	30.0	24.7	28.5	23.6	28.3	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.0	27.4	29.8	24.0	20.2	25.9	22.5	18.5	25.2	100.5	29.6	79.1	29.0	73.9	27.8	36.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	8.0	7.2	0.4	41.8	1.3	1.3	-0.5	42.8	-1.3	43.5	-2.0	44.3	-3.0	45.2		
(5)			-1.7	25.2	1.5	1.3	-2.7	26.1	-3.6	26.9	-4.4	27.6	-5.4	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	29.3	28.8	27.2	22.8	17.8	14.0	14.2	16.1	20.9	24.4	27.1	28.8
	DBStd	6.40	2.64	2.32	2.78	3.06	3.19	2.99	2.80	3.21	3.74	3.44	3.09	3.11
	HDD10.0	3	0	0	0	0	0	2	1	0	0	0	0	0
	HDD18.3	424	0	0	0	4	49	136	131	84	17	2	1	0
	CDD10.0	4592	597	526	533	385	242	121	131	190	327	448	512	582
	CDD18.3	1974	339	293	274	140	32	5	3	17	94	191	262	324
	CDH23.3	27122	4738	3986	3551	1682	402	78	110	385	1402	2675	3564	4549
	CDH26.7	15101	2856	2334	1966	749	99	13	9	106	639	1479	2076	2773
Wind	WSAvg	3.5	3.9	3.8	3.7	3.6	3.1	2.9	3.0	3.4	3.6	3.7	3.8	3.8
Precipitation	PrecAvg	276	64	55	18	18	23	10	9	5	9	18	25	56
	PrecMax	425	248	317	115	128	129	48	37	42	87	68	98	218
	PrecMin	78	0	0	0	0	0	0	3	0	0	0	0	5
	PrecStd	89	59	77	30	33	35	14	9	10	19	19	22	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	40.2	39.2	35.6	31.3	28.6	27.9	32.0	36.0	38.6	39.8	42.0
		MCWB	19.5	18.6	19.3	18.0	16.1	17.9	14.5	16.2	17.4	16.4	17.7	18.6
	2%	DB	39.6	38.5	37.5	33.8	29.2	25.5	26.0	29.4	33.9	37.1	38.4	40.1
		MCWB	19.2	18.7	19.3	17.4	15.1	14.4	13.2	15.3	16.4	16.4	17.5	18.4
	5%	DB	38.3	37.4	36.2	32.5	27.5	23.3	24.5	27.4	32.4	35.7	37.1	38.4
		MCWB	19.7	18.9	19.6	17.2	14.6	12.7	12.6	14.1	15.9	16.1	17.7	18.7
	10%	DB	36.9	36.2	34.8	31.1	25.6	21.3	22.5	25.3	30.7	34.0	35.6	36.9
		MCWB	19.6	19.0	19.2	17.0	14.4	11.6	11.6	12.8	15.0	16.0	17.3	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	32.1	34.4	30.7	22.9	19.8	17.6	26.2	29.5	24.8	22.9	24.9
		MCDB	30.6	32.6	34.9	31.1	23.0	21.5	22.5	26.8	30.4	27.6	31.0	29.5
	2%	WB	24.8	24.4	29.3	24.1	18.5	16.7	15.6	19.2	19.7	20.5	21.7	23.4
		MCDB	29.4	28.8	30.2	24.7	24.3	21.8	22.9	22.0	25.6	27.1	30.0	28.7
	5%	WB	23.9	23.5	23.6	20.0	16.6	14.8	13.7	15.7	17.5	19.3	20.8	22.7
		MCDB	28.5	28.4	27.7	25.1	24.0	20.4	22.4	24.3	27.4	26.9	28.7	28.6
	10%	WB	23.2	22.8	22.1	18.2	15.2	13.0	12.3	13.7	15.9	18.1	20.0	22.1
		MCDB	28.6	28.2	27.6	27.6	23.7	20.0	20.7	24.4	28.3	27.8	28.4	28.9
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.1	13.3	14.6	14.2	14.5	16.0	16.7	16.5	16.2	14.7	13.6
		MCDBR	15.5	15.5	15.4	16.2	16.4	16.4	18.3	19.2	18.3	19.0	16.9	16.2
		MCWBR	4.6	4.9	6.0	6.1	7.0	7.8	8.6	9.7	7.6	7.1	5.7	5.1
	5% WB	MCDBR	10.0	10.0	11.1	12.1	12.8	13.4	15.7	15.7	14.0	13.1	12.4	11.3
MCWBR		2.8	3.6	7.8	8.5	6.7	7.4	7.6	11.0	8.1	5.4	4.2	3.4	
Clear-Sky Solar Irradiance	taub		0.360	0.356	0.340	0.318	0.303	0.286	0.287	0.298	0.351	0.358	0.366	0.367
	taud		2.544	2.560	2.582	2.592	2.591	2.620	2.594	2.528	2.405	2.412	2.454	2.503
	Ebn at Noon		984	974	964	946	921	921	930	952	938	961	971	978
	Edh at Noon		111	107	100	92	84	78	83	96	117	123	120	115
All-Sky Solar Radiation	RadAvg		7.43	7.23	6.56	5.81	4.77	4.35	4.64	5.62	6.49	7.19	7.42	7.28
	RadStd		0.46	0.50	0.48	0.41	0.43	0.27	0.37	0.25	0.38	0.49	0.49	0.49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page